

# Design of the Main Drive Shaft for a Vertical Axis Hydrogenerator

MIET2510: Mechanical Design

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# Nomenclature

$A_s$ : Tensile stress area

$c_{-}$ : Maximum distance from neutral axis to the point, subscript as v: vertical, h: horizontal (mm)

$c$ : center to center of 2 sprockets (mm)

$d$ : Largest diameter of the bolt, crest diameter (mm)

$F_{wind}$ : Force caused by wind

$F_{tension}$ : Tension force applied on the bolts

$I$ : Area moment of inertia ( $m^4$ )

$L_{10}$ : Basic life (Rating life) of parts in hours, refers to the number of revolutions (or hours at a uniform speed) that 90% of the group of identical bearings will complete or exceed before the first evidence of failure develops

$L$ : length of the chain to connect 2 sprockets together (mm)

$M_{max}$ : maximum bending moment when wind force applied on the panel

$M_{Max}$ : Maximum bending moment applied on the system in gusting wind scenario

$p$ : Pitch of the bolt, distance between 2 crests

$r$ : radius of cross-section area, subscript as v: vertical, h: horizontal (mm)

$r_i$ : inner radius of the shaft, subscript as v: vertical, h: horizontal (mm)

$r_o$ : outer radius of the shaft, subscript as v: vertical, h: horizontal (mm)

$r_1, r_2$ : radius of the driving and the driven sprockets respectively (mm)

$S.F$ : Safety factor

$T_{rated}$ : required torque to orient the shaft

$T_{motor}$ : Rated output torque of the motor + slew drive

$V$ : Shear force applied on the bolts

$\alpha$ : contact angle

$\rho$ : radial distance, distance between  $\tau$  to center point (mm)

$\sigma_{allow}$ : allowable normal stress (MPa), used in calculating bending moment

**$\sigma_{\text{yield}}$** : yield strength (Mpa), used in calculating bending moment

**$\sigma_t$** : Tensile stress of the bolt

**$\tau$** : Shear stress of the bolt

**$\tau_{\text{max}}$** : maximum shear stress, in torsional deflection

# Abstract

The report presents the conceptual design of a dual-axis solar tracking system with given specifications for the solar panel array. By going through this report, a reliable system is designed in consideration for a load life of 8 years and external loadings like weights and gusting wind. Its functionality is to be able to rotate and tilt to maximize solar intake. Key elements like motors, bearings, and structural support are designed with detailed considerations through torque, and forces calculations. The system is designed with a safety factor of 2 and ensures 98% reliability. Also, lightweight, practicality, and reasonability are factors to be accounted for. After that, cost and vendors are suggested, ready for prototyping process. Some suggestions are included for future improvement.

## 1. Introduction

### 1.1. Background

Solar energy is a leading source of energy in the renewable energy field. But efficiency is dependent on solar availability and the orientation of photovoltaic (PV) panels relative to the sun. To maximize the solar power intake, a dual-axis solar tracking system allows the panel to follow the sun path by rotation and tilt. There are challenges like balancing structural strength and environmental resistance like wind gusts and corrosion.

### 1.2. Objectives

The designed solar tracking system needs to support and orient a  $2.5 \text{ m} \times 4.7 \text{ m}$ , 120kg photovoltaic panel to rotate and tilt to maximize solar exposure throughout the day. The minimum operational time is 8 years with a minimum reliability target of 98% and safety factor of 2 for all structural and load-bearing components. The system must include a vertical support column with a rotating shaft, a tilt mechanism, bearings, and a reliable structural frame. Motors and power transmission components should be considered to ensure sufficient torque for panel movement at a low but steady speed and resist shock loading caused by environmental forces such as gusting wind while keeping the system's weight and cost as reasonable as possible.

### 1.3. Scope

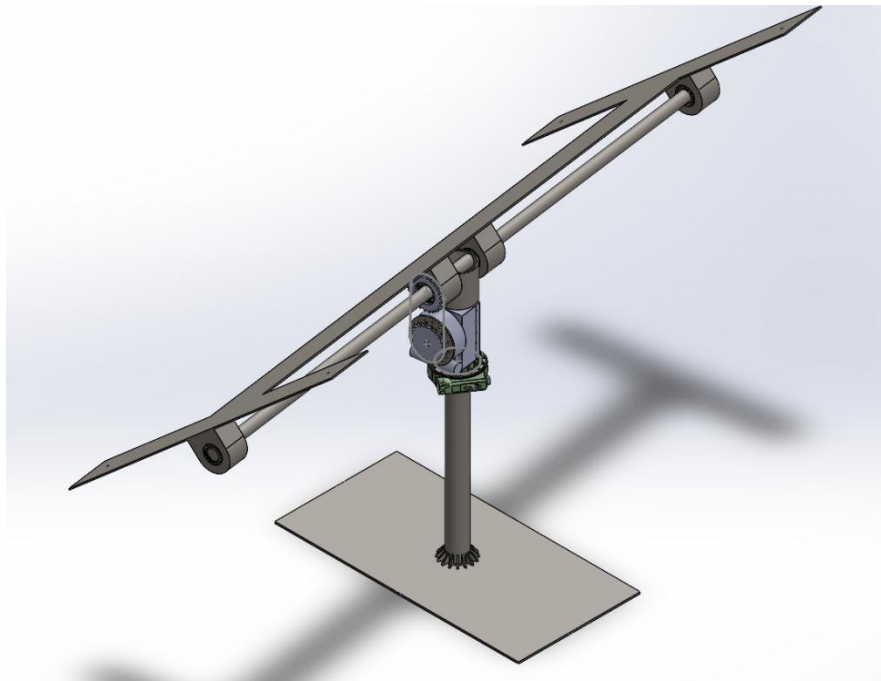
This design project only covers the mechanical side of the solar tracking system meaning elements to be selected include shafts, bearings, supporting structures, couplings, gears and drive motors. This includes stress analysis, material considerations to meet the requirements. The CAD 3D files are just a means of visualization and some parts are not 100% the same as in the real world.

## 1.4. Report structure

This report begins with an overview of the solar tracking system, then, after stating the assumptions, comes to the detailed load and torque calculations. Material selection, bearing and motor specifications, and power transmission design are then discussed. Structural considerations, including stability under wind loading, are presented, and conceptual CAD layouts are provided. The report concludes with a summary of key findings and recommendations for further refinement or testing.

# 2. Design Overview

The system will have one vertical support connect directly to the gear motor to rotate. All screws used in this design are M14 for easy manufacturing and installation. Another motor is installed on the vertical support, attached to the sprockets and chains linked to the panel holder for tilt control. 4 spherical roller bearings are used to support radial loadings from the panel. The outer surface of the support is coated with an oxidized layer to prevent corrosion. All components are chosen in cast iron steel for its strength and reasonable cost.



*Figure 1: Isometric view of solar tracking system*

## 2.1. Assumptions and Selections

Some common information will be assumed for easy calculations and component selections (detailed explanation will be put in Appendix 1 and 2).

Table 1: Assumption of speed, and hours over 8 years

| Assumptions                    | Value                  |
|--------------------------------|------------------------|
| Solar rotational speed (rad/s) | $7.292 \times 10^{-5}$ |
| Solar tilting speed (rad/s)    | $3.086 \times 10^{-5}$ |
| Load life targeted (hrs)       | 280,704                |

### 2.1.1. Material

| Properties                                    | Carbon Steels | Alloy Steels | Stainless Steels | Tool Steels |
|-----------------------------------------------|---------------|--------------|------------------|-------------|
| Density (1000 kg/m <sup>3</sup> )             | 7.85          | 7.85         | 7.75-8.1         | 7.72-8.0    |
| Elastic Modulus (GPa)                         | 190-210       | 190-210      | 190-210          | 190-210     |
| Poisson's Ratio                               | 0.27-0.3      | 0.27-0.3     | 0.27-0.3         | 0.27-0.3    |
| Thermal Expansion (10 <sup>-6</sup> /K)       | 11-16.6       | 9.0-15       | 9.0-20.7         | 9.4-15.1    |
| Melting Point (°C)                            |               |              | 1371-1454        |             |
| Thermal Conductivity (W/m-K)                  | 24.3-65.2     | 26-48.6      | 11.2-36.7        | 19.9-48.3   |
| Specific Heat (J/kg-K)                        | 450-2081      | 452-1499     | 420-500          |             |
| Electrical Resistivity (10 <sup>-9</sup> W-m) | 130-1250      | 210-1251     | 75.7-1020        |             |
| Tensile Strength (MPa)                        | 276-1882      | 758-1882     | 515-827          | 640-2000    |
| Yield Strength (MPa)                          | 186-758       | 366-1793     | 207-552          | 380-440     |
| Percent Elongation (%)                        | 10-32         | 4-31         | 12-40            | 5-25        |
| Hardness (Brinell 3000kg)                     | 86-388        | 149-627      | 137-595          | 210-620     |

Figure 2: Comparison chart of types of steels [3]

The center support will be under tensile and torsional loadings caused by the solar panel pressing down and the rotation of chain. The material that can withstand large force is Carbon Steel, but the system will be established outside, and corrosion is a factor affecting the lifetime of the system. That's why we will coat an oxidized layer to protect from corrosion.

### 2.1.2. Shaft dimensions

- Vertical shaft



With the effect caused by the shock loading like gusting winds calculated in appendix 3 known the maximum moment of 5034.66Nm, the minimum dimension of the support would be:

$$\sigma_{allow} = \frac{M_{max}}{I_{min}} c_v = \frac{M_{max}}{S_v} \rightarrow S_h = \frac{M_{max}}{\frac{\sigma_{yield}}{S.F}} = \frac{5034.66}{\frac{186 \times 10^6}{2}} = 5.414 \times 10^{-5}$$

$$S_v = \frac{I_v}{c_v} = \frac{\frac{\pi r_v^4}{4}}{r_v} = \frac{\pi r_v^3}{4} \rightarrow r_v = \sqrt[3]{\frac{4S_v}{\pi}} = 0.041m$$

This makes the minimum radius of the shaft resist the wind to be 0.041m. Hence The minimum diameter is 0.082m. The chosen diameter needs to be rechecked with the torsional deflection to ensure proper work.

#### - Horizontal shaft

For the tilting part, the panel holder will rotate around the horizontal shaft, making it the main support of the solar panel and panel holder. This horizontal support needs to overcome bending deflection caused by the wind.  $M_{max} = F_{wind} \times \frac{4.7}{2} = 3943.817Nm$ .

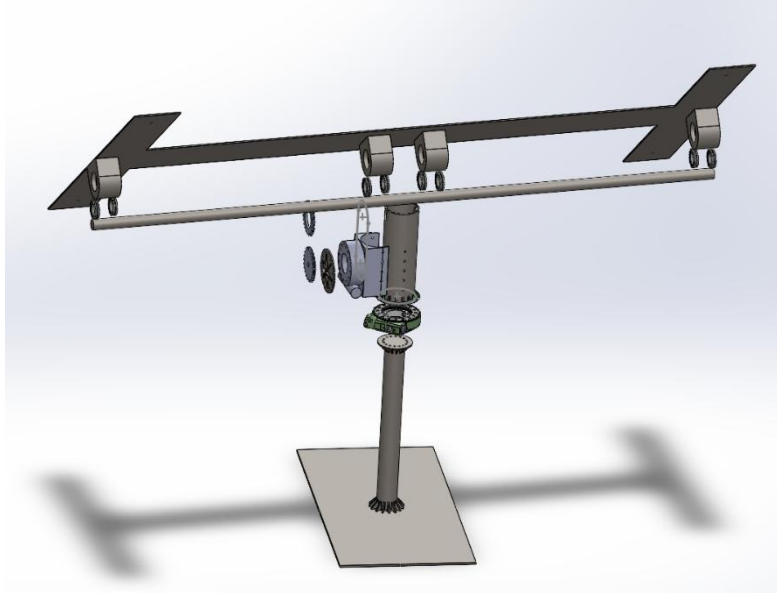
$$\sigma_{allow} = \frac{M_{max}}{I_{min}} c_h = \frac{M_{max}}{S_h}$$

$$S_h = \frac{M_{max}}{\sigma_{allow}} = \frac{3943.817}{\frac{186 \times 10^6}{2}} = 4.241 \times 10^{-5} = \frac{I_h}{c_h} = \frac{\frac{\pi r_h^4}{4}}{r_h} = \frac{\pi r_h^3}{4}$$

$$\rightarrow r_h = \sqrt[3]{\frac{4S_h}{\pi}} = 0.038m$$

To support the wind, the horizontal shaft is designed to have a diameter of 80 mm, carbon steel.

## 2.2. Exploded view



*Figure 3: Exploded view of the system*

Above is the exploded view of the whole system clearly depicting all components used in the conceptual design. From the top down is panel holder, bearings, horizontal shaft, vertical shaft, motor for tilt control and another horizontal shaft to link to 2 sprockets, motor for rotational control and the base. 2 gearmotors are connected to the slew drive. For detailed 2D drawings, refer to the Appendix.

It is advised that the motors are mounted first, then the bearings, next up is assembling the horizontal shaft to go through the bearings then the vertical shaft. For more stability, weld the horizontal shaft to the vertical shaft. The sprocket connected to the panel holder by welded. Make sure to keep a clearance of 5mm for the chain plate links.

## 3. Components Selection

### 3.1. Motor

To move the shaft of the support with a velocity of  $7.292 \times 10^{-5}$  rad/s, which is  $6.963 \times 10^{-4}$  rpm with a minimum torque of 4706.842 Nm needed (appendix 4). The market is investigated to find suitable motors for the system. Noted, the minimum power required is  $P_{min} = T \times \omega = 0.096W$ . Industrial motors with high torque at low speed are either servo or stepper motors. But these types of motor cannot lock itself in the no-power condition and external braking and worm gear are required to hold the panel in place. After some research, the industrial motor with built-in brake mechanism, having high torque with low-speed output, and widely used for solar system is Slew Drive SE9, bought from IMO, part number: WD-L 0223/3-04698. With the rated output torque of 4.795kN, holding torque of 9.303kN. When used with 24V DC gear motor and went through the 62:1 ratio of the slew drive, the output speed is

0.08rpm [4]. But this speed is still far too fast for accurate solar tracking. The solution is to update the position once every 60 seconds and lower the input speed from the gearmotor, meaning letting the motor run for 5 seconds every 10 minutes.

For the tilting motion, the same DC gearmotor is used. To match with  $3.086 \times 10^{-5}$  rad/s, or  $2.947 \times 10^{-4}$  rpm, the motor will run for 5 seconds every 23 minutes. To support the motor to level vertically, 2 supports are added on the sides.

### 3.2. Shaft

- From the minimum dimensions calculated above, the suitable shaft diameter can be chosen to fit with the chosen motor

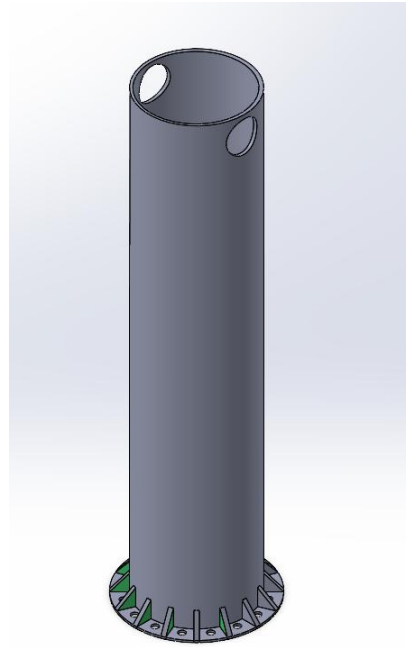


Figure 4: Isometric view of the vertical shaft

- Vertical shaft:

The vertical shaft, also act as a support will have the outer diameter of the support at the bottom fit with the outer diameter of the motor (325mm), the inner diameter will be 224 mm. Based on the  $S_h = \frac{I}{c} = 5.414 \times 10^{-5}$ , calculate the outer part to withstand the

weight and forces from the wind.  $\frac{I}{c} = \frac{\frac{\pi(r_o^4 - r_i^4)}{4}}{r_o} \rightarrow r_i = \sqrt[4]{r_o^4 - \frac{S_h 4 r_o}{\pi}} = 112 \text{ mm}$ . From the torsional shear stress equation,  $\tau_{max} = \frac{T}{J} \rho = \frac{T_{rated}}{\frac{\pi}{2}(r_o^4 - r_i^4)} r_o$

$$\rightarrow r_i = \sqrt[4]{r_o^4 - \frac{T_{rated} r_o}{\frac{\pi}{2} \tau_{max}}} = 112 \text{ mm.}$$

1mm thick wall is unreasonable, so 24 mm thick is chosen instead, making the outer wall of the shaft to be 248 mm. The requirement also states that the panel must be minimum 700mm above ground at maximum tilt. That

brings the height of the vertical shaft to  $\frac{2.5}{2} \times 10^{-3} \sin 40^\circ + 700 - 108 =$

1395.485mm. While 108mm is the height of the slew drive. To ensure it in the allowed torque range, the slew drive is put in the middle of the support, 700mm away from the panel holder.

#### - Horizontal shaft

The minimum diameter of a horizontal shaft to resist bending from wind is 80mm (calculated in 2.1.2).

Another horizontal shaft is connected to the motor. The dimensions are constrained by both the motor and the sprocket. The outer diameter will be 12mm.

### 3.3. Bearing

For the bearing in the support for the panel, it's mainly affected by radial load caused by the pole and the panel's gravity. Roller bearing is preferred here as it's good at handling radial loads. Also, during installation, there might exist some axial loadings. That's why **Spherical Roller Bearing** is chosen for this system. This type can withstand both radial and axial loadings, durable under shock, and can handle misalignment. Although this bearing has higher friction and is more expensive, it is still a good tradeoff for this system. The same applied to the bearing supporting the horizontal shaft and the panel holder.

Another thing to consider when choosing bearings is the maximum rotational speed. As per the calculations and assumption above (prefer to Assumptions in design overview), the shaft's rotation speed would be  $7.292 \times 10^{-5}$  rad/s. After calculating the load life (appendix 6), it's guaranteed to last at least 8 years.

For the horizontal shaft and the panel holder, the inner diameter needs to fit with the shaft, which is 80mm. As Spherical roller bearing is good for both radial and axial loads, it is also suitable for this position. Hence, from SKI standard, choose model SKF **22216E (80 × 140 × 33 mm)**[5].

### 3.4. Chain sprockets

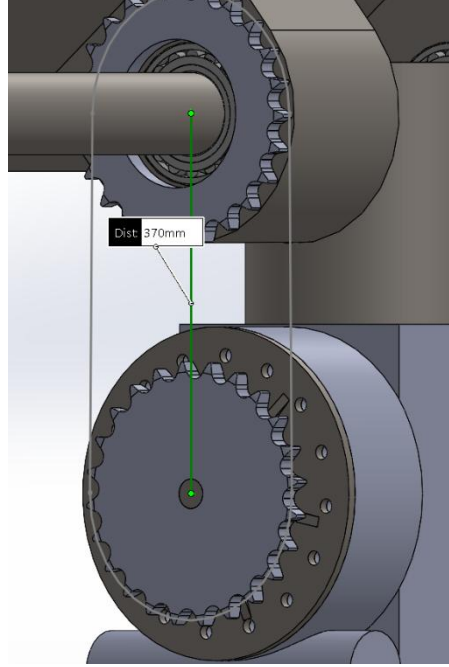


Figure 5: Chain and sprockets in 3D solidworks

When comparing chain to belts and gears, it has some distinctive advantages like longer load life, more accurate and can transmit power over a long distance.

All the speed control of the tilting motion are left to the slew drive to have the desired output speed of  $2.947 \times 10^{-4} \text{ rpm}$ . Hence, the ratio of 2 chain sprockets is 1:1. The dimensions of the sprockets will depend on the diameter of the motor and the availability in the market.

At first, a sprocket with an inner diameter of 12 mm and 24 teeth is chosen. With the same pitch and numbers of teeth, the sprocket connecting to the horizontal shaft is drilled to be 85 inner diameters (5 mm away from the static shaft) to leave space for welding. The outer diameter is 104.14mm  $\rightarrow r_1 = r_2 = 52.07 \text{ mm}$ . One sprocket is connected to the motor, positioned 700 mm away from the panel holder  $\rightarrow c \cong 370 \text{ mm}$  measured in solidworks.

The motor creates a rated torque of 4.795kNm. The force applied on the chain would be  $F = \frac{T_{motor}}{r_1} = \frac{4.795 \times 10^3}{52.07 \times 10^{-3}} = 92 \text{ kN}$ . With S.F of 2, the chain and sprockets need to withstand at least 184kN. Very high force applied on the chain can reduce operational life.

That's why, another option needed to adopted, which is to use 2 strands 20B chain with 20B pitch sprockets with diameter of 243 mm. The chain tension reduced to 39.43kN. Maximum allowable (fatigue) load for a single 20B strand  $\approx 18.4 \text{ kN}$ , so duplex allowable  $\approx 36.8 \text{ kN}$ . The manufacture datasheet states that the Tensile Strength to be 95N [6] which is closely satisfied the required strength of the chain.

Take the same c, recalculate with new radius of 121.5 mm, the length of the chain will be

$$L = 2\sqrt{c^2 - (r_2 - r_1)^2} + r_1(\pi - 2\alpha) + r_2(\pi + 2\alpha)$$

$$L = 2\sqrt{370^2} + 2 \times 121.5\pi = 1503.407 \text{ mm}$$

### 3.5. Screw

Table 2: Mechanical and physical properties as per ISO 898-1 for Class 8.8 (M14)[7]

| Mechanical or Physical Property                          | Class 8.8 (M16 and below) |
|----------------------------------------------------------|---------------------------|
| Tensile strength (Mpa)                                   | 800                       |
| Stress at 0.2% non-proportional elongation (yield) (Mpa) | 640                       |
| Stress under proof load (Mpa)                            | 580                       |

The external forces caused by the wind  $F_{wind} = 1678.22N$ ;  $M_{max} = 5034.66Nm$ . With 16 bolts used at every joint, the tension  $F_{tension} = \frac{M_{max}}{16r_v} = 1.936kN$ , the shear force  $V = \frac{F_{wind}}{16} = 104.889N$

M14 with standard  $p=2mm$ ,  $d=14mm \rightarrow A_s = 4\pi(d - 0.9382p)^2 \approx 115.44 \text{ mm}^2$

The stresses produced  $\sigma_t = \frac{F_{tension}}{A_s} = 16.77MPa$ ;  $\tau = \frac{V}{A_s} = 0.908MPa$ . It is safe to use these bolts as such.

For the vertical mounted parts, the forces applied are weight caused by the motor and slew drive:  $V = \frac{50 \times 9.81}{10} = 49.05N$ . For 2 bolts holding the slew drive and the support together:  $V = \frac{50 \times 9.81}{2} = 245.25N \rightarrow \tau = 2.124MPa$ . Still functionable.

## 4. Result

In summary, followings are all the parts and their cost (estimated) and build of materials.

Table 3: Build of material for the system

| Component                    | Description                                           | Model                                    | Quantity | Cost (estimated) USD |
|------------------------------|-------------------------------------------------------|------------------------------------------|----------|----------------------|
| Motor                        | rated output torque=4.795kN<br>holding torque=9.303kN | Slew Drive SE9 with 24VDC gear motor [8] | 2        | \$300*2              |
| Bearing (horizontal support) | spherical roller bearing                              | SKF 22216 E (80 × 140 × 33 mm)           | 8        | \$263.49*8[9]        |

|           |                                        |     |    |                |
|-----------|----------------------------------------|-----|----|----------------|
|           | 80 mm bore<br>140 mm outer<br>diameter |     |    |                |
| Screw     | M14 (110*14)                           | M14 | 16 | \$4*16[10]     |
| Screw     | M14 (125*14)                           | M14 | 14 | \$48.1 [11]    |
| Screw     | M14 (80*14)                            | M14 | 10 | \$0.68*10 [12] |
| Nut       | Lock nut                               | M14 | 56 | 1.29*56 [13]   |
| Chain     | 10ft, 20B                              | 20B | 1  | \$504.99 [6]   |
| Sprockets | 24 teeth                               | 20B | 2  | \$42.59*2 [15] |

For the dimensions of the shafts and other structural details, it is advised to refer to Appendix for detailed 2D drawing.

*Table 4: Summary of dimensions of every shaft*

| Part name        | Diameter (mm) | Height (mm) |
|------------------|---------------|-------------|
| Vertical shaft   | 224*248       | 700         |
| Base             | 147           | 1310        |
| Horizontal shaft | 80            | 4700        |
| Horz shaft       | 12*325        | 50          |

## 5. Discussion

With calculated lifetime of components, the system has higher reliability. The torque calculated to be in range of motor's functionality. Having vertical support with sized bearing and carefully considered components like chains, sprocket and structures with dimensions followed a safety factor of 2, ensuring resilience against gusting wind loads and shock conditions. Overall, the design demonstrates a practical balance between durability and economic feasibility. The conceptual design utilizes many industrial standard components, reflecting industry practicality.

This system still has several problems in assumptions in environmental effects to only have wind gusts while ignoring the possibility of corrosion, fatigue from thermal expansion. The design doesn't account for manufacturing tolerances, potential misalignments, or dynamic loading during start-up and braking conditions. While the safety factor of 2 mitigates these uncertainties, further detailed testing and simulations would be necessary to fully validate the design.

## 6. Conclusion and future improvements

### 4.1. Conclusion

Through the design process, detailed, off-the-shelf components are chosen to complete the conceptual design of the dual-axis solar tracking system able to support and orient a photovoltaic

array of 120 kg with dimensions of 2.5 m × 4.7 m. The design incorporated critical mechanical elements, including a vertical support column, motorized rotation and tilt mechanisms, and structural reinforcement to ensure both functionality and durability under static and dynamic loads.

Through torque and load calculations, motors are chosen to provide reliable low-speed operation with sufficient safety margin against environmental wind loading. Bearings withstand axial and radial loads to ensure long life and stability of the panel. Power transmission was achieved using a chain and sprocket arrangement, chosen for its robustness, ease of maintenance, and efficiency at low rotational speeds. The structural frame dimensions were optimized to maintain ground clearance, stability, and resistance to shock loading, with all components designed to a factor of safety of 2. The final design satisfies the project requirements of long-term reliability (8-year life), operational safety, and cost-effectiveness while ensuring increased solar energy capture through precise sun tracking.

## **4.2. Future improvements**

- As the scope of this project stops at mechanical side, for future development, electronics side (inverter, power bank, etc.) and motor controlling (using driver and microcontroller) are to be considered.
- A solar radiation sensor can be implemented to control and accommodate with the Sun's change throughout the year, further improving the solar energy intake.
- Better material, structural design should be considered to increase the resistance against environmental factors such as wind and reduce the cost and weight of motors. By performing finite element analysis (FEA) for every segment, stress distribution in shafts, bolts will be more accurately addressed.
- A manual should be developed as many parts require lubrication such as chains and bearings. So, it is advised that occasional maintenance should be provided to ensure efficiency.
- Dust on the panel can be a problem if unattended in 8 years period. Suggestion is to have a vibration mechanism to shake off the dust, maintaining high efficiency.

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# Appendix

## Appendix 1. Solar speed calculation

- Rotational speed: The following assumption that this system is installed in Vietnam in August, when the earth rotational speed is  $7.292 \times 10^{-5}$  rad/s (around 15 degree per hour) [1] will be applied to the designing process of the system. That makes the angular speed around the pole's axis is  $7.292 \times 10^{-5}$  rad/s.
- Tilting speed: For the tilt of the panel, assume the sun rises at 5:43 and sets at 18:08 and the time the sun highest is at noon (12:00). That means the panel has to tilt from 40 deg to 0 deg in  $t = 12 - 5.7167 = 6.2833$  (hrs). Hence, the velocity would be
$$v = \frac{\theta}{t} = \frac{\frac{2\pi}{9}}{6.2833 \times 3600} = \frac{\frac{2\pi}{9}}{6.2833 \times 3600} = 3.086 \times 10^{-5} \text{ (rad/s).} [2]$$

## Appendix 2. Target load life calculation

In 8 years of the system's operation, leap year will occur twice. So, the number of hours to be taken into consideration is  $L_{10} = 8 \times (365 \times 4 + 2) \times 24 = 280,704$  (hrs).

## Appendix 3. Wind force calculation

Shock loading like gusting winds (25 knots = 12.86 m/s), wind pressure  $p = \frac{1}{2} \rho V^2 = \frac{1}{2} \times 1.225 \times 12.86^2 \cong 101 \text{ Pa}$ .

➔ The force acting on the panel (at 40° tilted):  $F = p C_d A$  ( $C_d$  is drag coefficient = 1.1)

The estimated force of the wind becomes  $F = 101 \times 1.1 \times 2.5 \sin 40^\circ \times 4.7 = 839.11 \text{ N}$

With safety factor of 2,  $F_{wind} = 2 \times F = 1678.22N$ .

➔ The maximum moment would be  $M_{max} = F \cdot d \cong 1678.22 \times 3 \cong 5034.66Nm$ .

Assumed the system would be around 3m tall.

#### Appendix 4. Torque required calculation

##### - Torque for rotating

The motor will be attached to a gear connected to the shaft. To start rotating (assume it starts from 0 to  $7.292 \times 10^{-5}$  rad/s in 1s), the minimum torque of the shaft needs to overcome static friction caused by the shaft, bearings and the whole system above (assumedly the weight of the panel and the above system), also the acceleration inertia. The shaft also needs to counter the wind flow (force caused by wind is calculated above  $F_{wind} = 1678.22N$ , applied on the panel's farthest point to be  $\frac{4.7}{2} = 2.35m$ :

$$T_{rated} = F_f \times r + I\alpha + F_{wind} \times 2.35$$

Where I is moment of inertia ( $kgm^2$ ) and  $\mu_s = 0.6$ [14]

$$\begin{aligned} T_{rated} &= F_f \times r + mr^2 \times \frac{\omega}{t} = \mu_s Nr + mr^2 \times \frac{\omega}{t} + F_{wind} \times 2.35 \\ &= 0.6 \times (270 + 120) \times 9.81 \times 0.1625 + (270 + 120) \times 0.1625^2 \times \frac{7.292 \times 10^{-5}}{1} \\ &\quad + 1678.22 \times 2.35 \\ &= 4706.842Nm \end{aligned}$$

##### - Torque for tilting

For rotating the panel, the torque needs to overcome the torque caused by wind and weight of panel holder only.

$$\begin{aligned} M_{max} &= m_{holder} \times 9.81 \times r + F_{wind} \times \frac{2.5}{2} = 270 \times 9.81 \times 0.040 + 1678.22 \times 1.25 \\ &= 2203.723N \end{aligned}$$

$$M_{tilt} = S.F \times M_{max} = 2 \times 2203.723 = 4407.446N$$

#### Appendix 5. Weight of panel holder

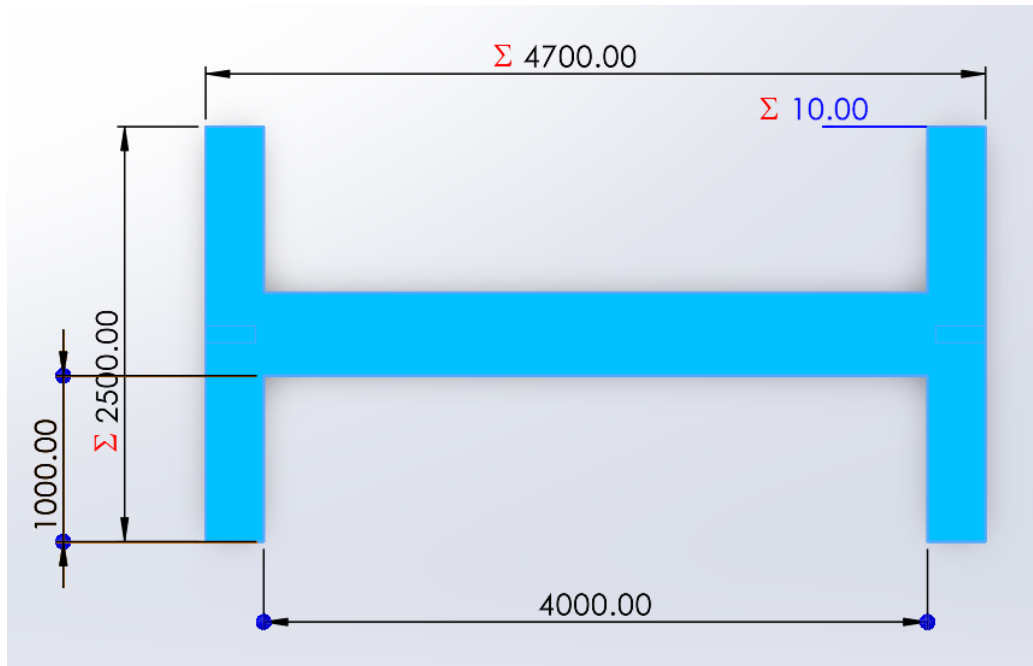


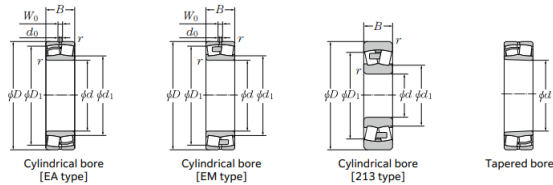
Figure 6: Dimensions of panel holder

Using the Steelexpress website and input in the width, length and depth as shown in the figure, the weight of cast steel panel holder is calculated as  $846kg - 2 \times 288 = 270kg$ . This is to estimate the weight applied on the motor.

## Appendix 6. Bearing load life

## Spherical Roller Bearings

NTN



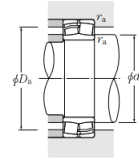
d 65 ~ 95mm

|    | Boundary dimensions |      |     |                                  | Basic load rating |                | Fatigue load limit |                 | Allowable speed |                                            | Bearing numbers <sup>1)</sup> |                  |
|----|---------------------|------|-----|----------------------------------|-------------------|----------------|--------------------|-----------------|-----------------|--------------------------------------------|-------------------------------|------------------|
|    | d                   | D    | B   | r <sub>0</sub> max <sup>3)</sup> | W <sub>0</sub>    | d <sub>0</sub> | C <sub>r</sub>     | C <sub>0e</sub> | C <sub>90</sub> | min <sup>-1</sup><br>Grease<br>lubrication | Oil<br>lubrication            | Cylindrical bore |
| 65 | 120                 | 31   | 1.5 | 8                                | 3.5               | 226            | 224                | 18.2            | 4 400           | 5 500                                      | *22213EAD1                    | *22213EAKD1      |
|    | 120                 | 31   | 1.5 | 8                                | 3.5               | 217            | 212                | 17.2            | 4 400           | 5 500                                      | *22213EMD1                    | *22213EMKD1      |
|    | 140                 | 33   | 2.1 | 7                                | 4                 | 216            | 228                | 31.0            | 3 100           | 4 000                                      | 21313                         | 21313K           |
|    | 140                 | 48   | 2.1 | 9                                | 4                 | 369            | 343                | 23.4            | 3 300           | 4 100                                      | *22313EAD1                    | *22313EAKD1      |
| 70 | 140                 | 48   | 2.1 | 9                                | 4                 | 369            | 343                | 23.4            | 3 300           | 4 100                                      | *22313EMD1                    | *22313EMKD1      |
|    | 125                 | 31   | 1.5 | 7                                | 3.5               | 235            | 240                | 20.1            | 4 100           | 5 200                                      | *22214EAD1                    | *22214EAKD1      |
|    | 125                 | 31   | 1.5 | 7                                | 3.5               | 217            | 212                | 17.2            | 4 100           | 5 200                                      | *22214EMD1                    | *22214EMKD1      |
|    | 150                 | 35   | 2.1 | 7                                | 4                 | 245            | 262                | 33.5            | 2 900           | 3 800                                      | 21314                         | 21314K           |
| 75 | 150                 | 51   | 2.1 | 10                               | 5                 | 420            | 396                | 26.0            | 3 000           | 3 800                                      | *22314EAD1                    | *22314EAKD1      |
|    | 150                 | 51   | 2.1 | 10                               | 5                 | 420            | 396                | 26.0            | 3 000           | 3 800                                      | *22314EMD1                    | *22314EMKD1      |
|    | 130                 | 31   | 1.5 | 7                                | 3.5               | 244            | 249                | 21.1            | 4 000           | 5 000                                      | *22215EAD1                    | *22215EAKD1      |
|    | 130                 | 31   | 1.5 | 7                                | 3.5               | 244            | 249                | 21.1            | 4 000           | 5 000                                      | *22215EMD1                    | *22215EMKD1      |
| 80 | 160                 | 37   | 2.1 | 7                                | 4                 | 266            | 287                | 27.5            | 2 700           | 3 500                                      | 21315                         | 21315K           |
|    | 160                 | 55   | 2.1 | 10                               | 5                 | 491            | 467                | 29.8            | 2 900           | 3 600                                      | *22315EAD1                    | *22315EAKD1      |
|    | 160                 | 55   | 2.1 | 10                               | 5                 | 491            | 467                | 29.8            | 2 900           | 3 600                                      | *22315EMD1                    | *22315EMKD1      |
|    | 140                 | 33   | 2   | 8                                | 3.5               | 278            | 287                | 24.0            | 3 700           | 4 600                                      | *22216EAD1                    | *22216EAKD1      |
| 85 | 140                 | 33   | 2   | 8                                | 3.5               | 267            | 272                | 22.8            | 3 700           | 4 600                                      | *22216EMD1                    | *22216EMKD1      |
|    | 170                 | 39   | 2.1 | 7                                | 4                 | 289            | 315                | 30.5            | 2 500           | 3 300                                      | 21316                         | 21316K           |
|    | 170                 | 58   | 2.1 | 10                               | 5                 | 541            | 522                | 32.5            | 2 700           | 3 400                                      | *22316EAD1                    | *22316EAKD1      |
|    | 170                 | 58   | 2.1 | 10                               | 5                 | 541            | 522                | 32.5            | 2 700           | 3 400                                      | *22316EMD1                    | *22316EMKD1      |
| 90 | 150                 | 36   | 2   | 8                                | 3.5               | 324            | 330                | 27.1            | 3 400           | 4 300                                      | *22217EAD1                    | *22217EAKD1      |
|    | 150                 | 36   | 2   | 8                                | 3.5               | 324            | 330                | 27.1            | 3 400           | 4 300                                      | *22217EMD1                    | *22217EMKD1      |
|    | 180                 | 41   | 3   | 7                                | 4                 | 320            | 355                | 45.0            | 2 400           | 3 100                                      | 21317                         | 21317K           |
|    | 180                 | 60   | 3   | 11                               | 5                 | 599            | 604                | 36.4            | 2 600           | 3 200                                      | *22317EAD1                    | *22317EAKD1      |
| 95 | 180                 | 60   | 3   | 11                               | 5                 | 599            | 604                | 36.4            | 2 600           | 3 200                                      | *22317EMD1                    | *22317EMKD1      |
|    | 160                 | 40   | 2   | 10                               | 4.5               | 384            | 398                | 30.2            | 3 200           | 4 000                                      | *22218EAD1                    | *22218EAKD1      |
|    | 160                 | 40   | 2   | 10                               | 4.5               | 384            | 398                | 30.2            | 3 200           | 4 000                                      | *22218EMD1                    | *22218EMKD1      |
|    | 160                 | 52.4 | 2   | 9                                | 4                 | 467            | 513                | 30.0            | 2 600           | 3 200                                      | *22318EAD1                    | *22318EAKD1      |
| 95 | 190                 | 43   | 3   | 7                                | 4                 | 355            | 400                | 50.5            | 2 300           | 3 000                                      | 21318                         | 21318K           |
|    | 190                 | 64   | 3   | 12                               | 5                 | 668            | 652                | 40.0            | 2 500           | 3 000                                      | *22318EAD1                    | *22318EAKD1      |
|    | 190                 | 64   | 3   | 12                               | 5                 | 668            | 652                | 40.0            | 2 500           | 3 000                                      | *22318EMD1                    | *22318EMKD1      |
|    | 170                 | 43   | 2.1 | 10                               | 4.5               | 416            | 417                | 33.4            | 3 000           | 3 800                                      | *22219EAD1                    | *22219EAKD1      |
| 95 | 170                 | 43   | 2.1 | 10                               | 4.5               | 416            | 417                | 33.4            | 3 000           | 3 800                                      | *22219EMD1                    | *22219EMKD1      |

1) Bearing part numbers with \* are ULTAGE Series and have outer ring oil holes and oil grooves as standard.  
2) "K" indicates bearings having a tapered bore with a taper ratio of 1:12. 3) Smallest allowable dimension for chamfer dimension r.  
B-220

## Spherical Roller Bearings

NTN



Dynamic equivalent radial load

$$P_1 = X F_r + Y F_a$$

|                          |                       |
|--------------------------|-----------------------|
| $\frac{F_a}{F_r} \leq e$ | $\frac{F_a}{F_r} > e$ |
| $X = 1$                  | $X = Y$               |
| $Y = 0$                  | $Y = 1$               |

Static equivalent radial load

$$P_0 = F_r + Y_0 F_a$$

For values of e, Y, Y<sub>0</sub> and Y<sub>2</sub> see the table below.

| Installation-related dimensions |                    |                    |                | Constant            |      | Axial load factors |                |                |  | Mass (approx.)   |              |
|---------------------------------|--------------------|--------------------|----------------|---------------------|------|--------------------|----------------|----------------|--|------------------|--------------|
| d <sub>1</sub>                  | d <sub>a</sub> min | D <sub>a</sub> max | D <sub>1</sub> | r <sub>as</sub> max | e    | Y <sub>1</sub>     | Y <sub>2</sub> | Y <sub>0</sub> |  | Cylindrical bore | Tapered bore |
| 78                              | 74                 | 111                | 107            | 1.5                 | 0.24 | 2.79               | 4.15           | 2.73           |  | 1.52             | 1.49         |
| 78                              | 74                 | 111                | 107            | 1.5                 | 0.24 | 2.79               | 4.15           | 2.73           |  | 1.53             | 1.5          |
| 85                              | 77                 | 128                | 119            | 2                   | 0.25 | 2.69               | 4.00           | 2.63           |  | 2.55             | 2.51         |
| 81                              | 77                 | 128                | 122            | 2.1                 | 0.33 | 2.06               | 3.06           | 2.01           |  | 3.48             | 3.41         |
| 81                              | 77                 | 128                | 122            | 2.1                 | 0.33 | 2.06               | 3.06           | 2.01           |  | 3.5              | 3.43         |
| 84                              | 79                 | 116                | 113            | 1.5                 | 0.22 | 3.01               | 4.48           | 2.94           |  | 1.61             | 1.58         |
| 84                              | 79                 | 116                | 113            | 1.5                 | 0.22 | 3.01               | 4.48           | 2.94           |  | 1.64             | 1.6          |
| 91                              | 82                 | 138                | 126            | 2                   | 0.25 | 2.69               | 4.00           | 2.63           |  | 3.18             | 3.14         |
| 85                              | 82                 | 138                | 131            | 2.1                 | 0.34 | 2.00               | 2.98           | 1.96           |  | 4.25             | 4.16         |
| 85                              | 82                 | 138                | 131            | 2.1                 | 0.34 | 2.00               | 2.98           | 1.96           |  | 4.31             | 4.22         |
| 88                              | 84                 | 121                | 118            | 1.5                 | 0.22 | 3.14               | 4.67           | 3.07           |  | 1.67             | 1.64         |
| 88                              | 84                 | 121                | 118            | 1.5                 | 0.22 | 3.14               | 4.67           | 3.07           |  | 1.71             | 1.67         |
| 99                              | 87                 | 148                | 136            | 2                   | 0.24 | 2.84               | 4.23           | 2.78           |  | 3.81             | 3.76         |
| 91                              | 87                 | 148                | 139            | 2.1                 | 0.34 | 2.00               | 2.98           | 1.96           |  | 5.18             | 5.07         |
| 91                              | 87                 | 148                | 139            | 2.1                 | 0.34 | 2.00               | 2.98           | 1.96           |  | 5.27             | 5.16         |
| 94                              | 91                 | 129                | 127            | 2                   | 0.22 | 3.14               | 4.67           | 3.07           |  | 2.09             | 2.05         |
| 94                              | 91                 | 129                | 127            | 2                   | 0.22 | 3.14               | 4.67           | 3.07           |  | 2.11             | 2.07         |
| 105                             | 92                 | 158                | 144            | 2                   | 0.23 | 2.95               | 4.40           | 2.89           |  | 4.53             | 4.47         |
| 98                              | 92                 | 158                | 148            | 2.1                 | 0.34 | 2.00               | 2.98           | 1.96           |  | 6.12             | 5.99         |
| 98                              | 92                 | 158                | 148            | 2.1                 | 0.34 | 2.00               | 2.98           | 1.96           |  | 6.28             | 6.15         |
| 100                             | 96                 | 139                | 137            | 2                   | 0.22 | 3.07               | 4.57           | 3.00           |  | 2.59             | 2.54         |
| 100                             | 96                 | 139                | 137            | 2                   | 0.22 | 3.07               | 4.57           | 3.00           |  | 2.67             | 2.62         |
| 111                             | 99                 | 166                | 152            | 2.5                 | 0.25 | 2.69               | 4.00           | 2.63           |  | 5.35             | 5.28         |
| 107                             | 99                 | 166                | 157            | 3                   | 0.32 | 2.09               | 3.11           | 2.04           |  | 7.18             | 7.04         |
| 107                             | 99                 | 166                | 157            | 3                   | 0.32 | 2.09               | 3.11           | 2.04           |  | 7.29             | 7.15         |
| 105                             | 101                | 149                | 144            | 2                   | 0.23 | 2.90               | 4.31           | 2.83           |  | 3.34             | 3.27         |
| 105                             | 101                | 149                | 144            | 2                   | 0.23 | 2.90               | 4.31           | 2.83           |  | 3.43             | 3.37         |
| 104                             | 101                | 149                | 141            | 2                   | 0.30 | 2.25               | 3.34           | 2.20           |  | 4.43             | 4.31         |
| 119                             | 104                | 176                | 162            | 2.5                 | 0.24 | 2.84               | 4.23           | 2.78           |  | 6.3              | 6.21         |
| 110                             | 104                | 176                | 166            | 3                   | 0.33 | 2.06               | 3.06           | 2.01           |  | 8.42             | 8.25         |
| 110                             | 104                | 176                | 166            | 3                   | 0.33 | 2.06               | 3.06           | 2.01           |  | 8.53             | 8.35         |
| 110                             | 107                | 158                | 153            | 2.1                 | 0.23 | 2.95               | 4.40           | 2.89           |  | 3.98             | 3.9          |
| 110                             | 107                | 158                | 153            | 2.1                 | 0.23 | 2.95               | 4.40           | 2.89           |  | 4.06             | 3.98         |

Note: For the bearings other than ULTAGE Series, outer rings with oil inlets and oil grooves can also be made based on your request.  
In this case, supplementary suffix "D1" is added after a bearing number. Example: 21317D1  
B-221

Figure 7: Factors for Spherical roller bearings [5]

- For horizontal support: SKF 22216 E (80 × 140 × 33 mm) rotating on the outer ring  
→  $V = 1.2$

Equivalent Radial Load

$$F_r = (120 + 270) \times 9.81 = 3.826 \text{ kN}$$

$$F_a = F_{\text{wind}} = 1.678 \text{ kN}$$

$$\text{As } \frac{F_a}{F_r} = 0.439 > 0.22 \rightarrow X = 0.67, Y = Y_2 = 4.67$$

$$P = X V F_r + Y F_a = 10.912$$

Load life calculation for rotational speed  $n = 2.947 \times 10^{-4} \text{ rpm}$

$$L_{10} = \frac{10^6}{60n} \left( \frac{C}{P} \right)^{\frac{10}{3}} = \frac{10^6}{60 \times 2.947 \times 10^{-4}} \left( \frac{278}{10.912} \right)^{\frac{10}{3}} = 2.752 \times 10^{12} \text{ (hrs)}$$

Satisfied the requirement of 280,704 hrs.

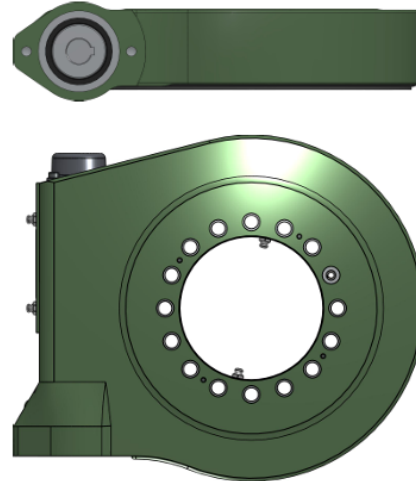
## Appendix 7. Datasheet of Slew drive

WD-L 0223/3-04698

3D View



2D derivation



[mm]

#### Technical data

|                                      |                              |
|--------------------------------------|------------------------------|
| Drawing number                       | WD-L 0223/3-04698            |
| Design size                          | 223                          |
| Max. operating torque / Nm           | 9303                         |
| Nom. operating torque / Nm           | 4795                         |
| Max. holding torque / Nm             | 9303                         |
| Static load rating, radial / kN      | 204                          |
| Static load rating, axial / kN       | 547                          |
| Dynamic load rating, radial / kN     | 132                          |
| Dynamic load rating, axial / kN      | 154                          |
| Ratio Slew drive                     | 62.00                        |
| Module ax./transverse section / mm   | 5                            |
| Number of threads of the worm        | 1                            |
| Number of teeth, wheel               | 62                           |
| Number of drives                     | 1                            |
| Type of gearing                      | 3 - Hardened tooth flanks    |
| Slewing ring design                  | Single Row Ball Slewing Ring |
| Mounting holes inner ring/worm wheel | Threaded holes               |
| Radial clearance / mm                | 0 - Preloaded                |
| Axial tilting clearance / mm         | 0 - Preloaded                |
| Length / mm                          | 356.0                        |
| Width / mm                           | 421.0                        |
| Height / mm                          | 100.0                        |
| Weight $\pm 10\%$ / kg               | 44                           |

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Figure 8: Datasheet of WD-L 0223/3-04698 slew drive

## Appendix 8. 2D drawings of every part in the system

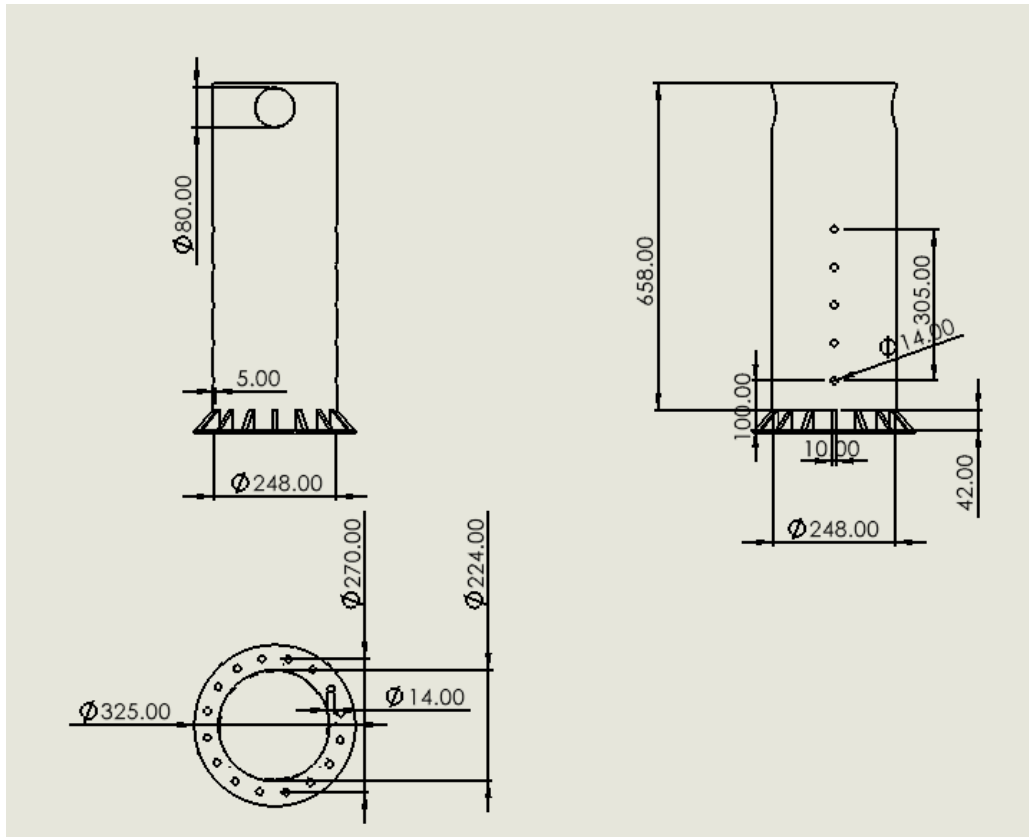


Figure 9: 2D drawings of vertical shaft

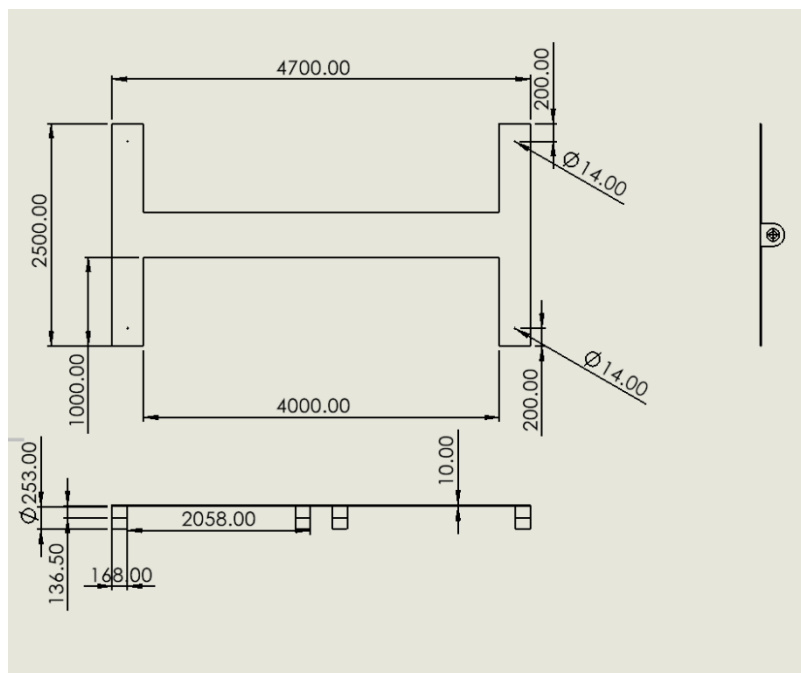
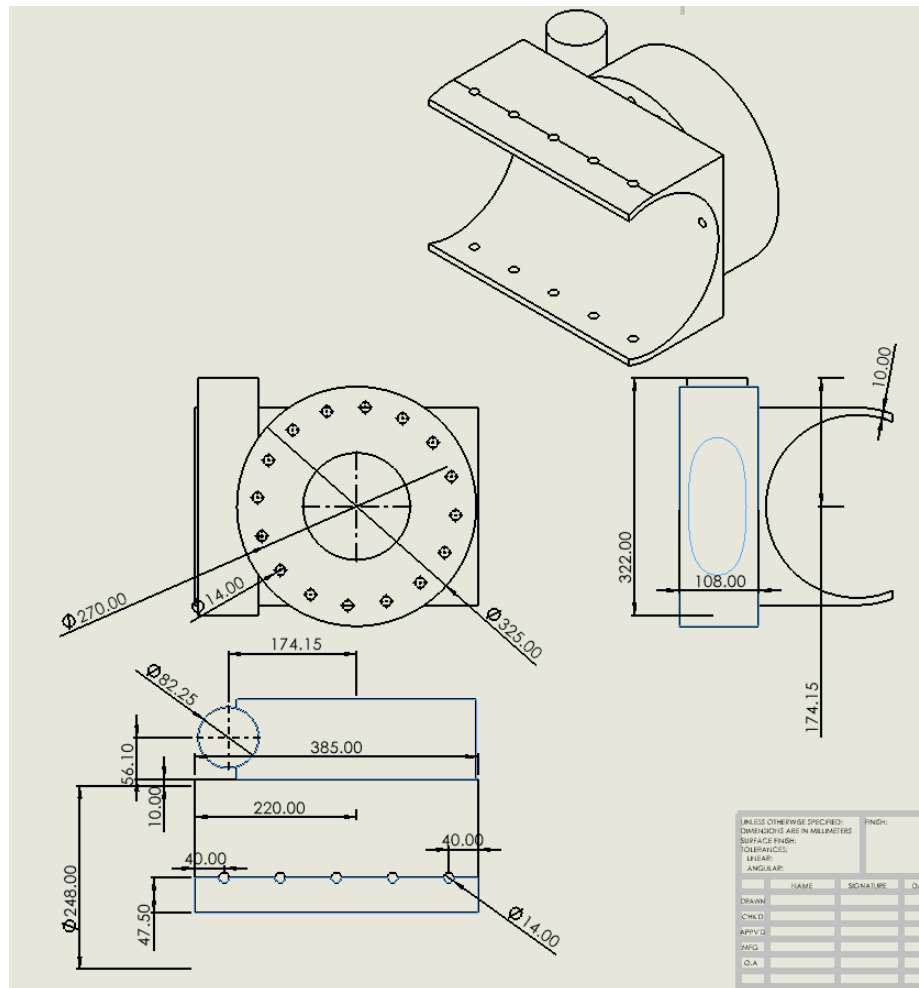


Figure 10: 2D drawing of panel holder



Note that this is not accurate, just treat as a model for required dimensions like diameters and holes for fastening. A 24VDC gearmotor will connect to the slew drive. The “C” part is to mount onto the vertical shaft.



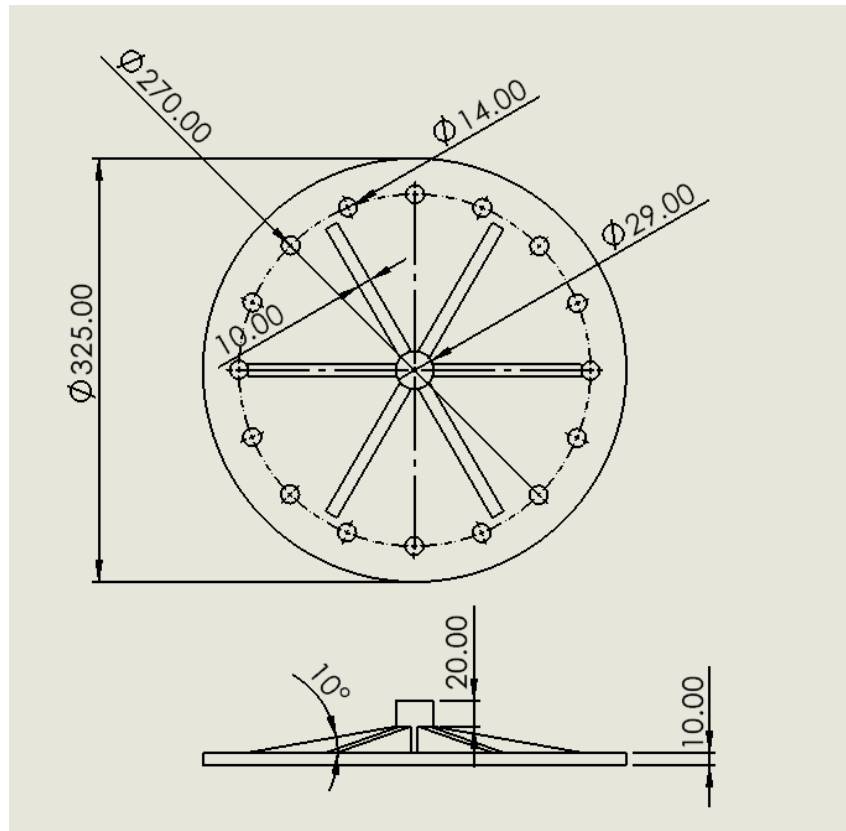


Figure 12: 2D drawing of horizontal shaft connected to driving sprocket

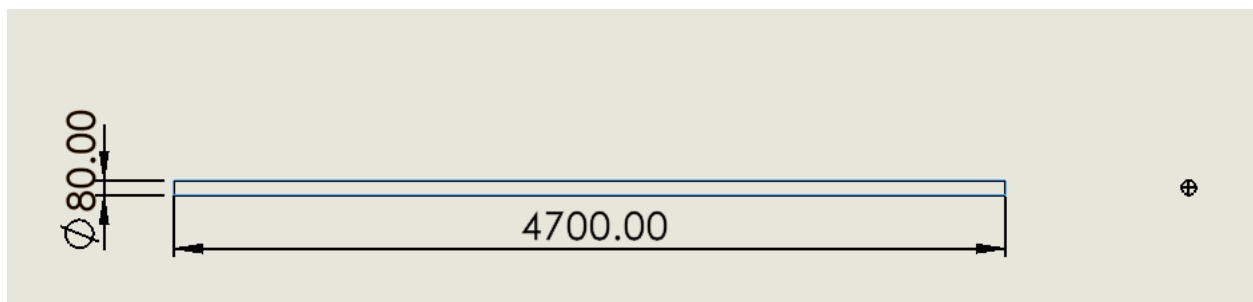


Figure 13: 2D drawing of horizontal shaft supporting the panel holder

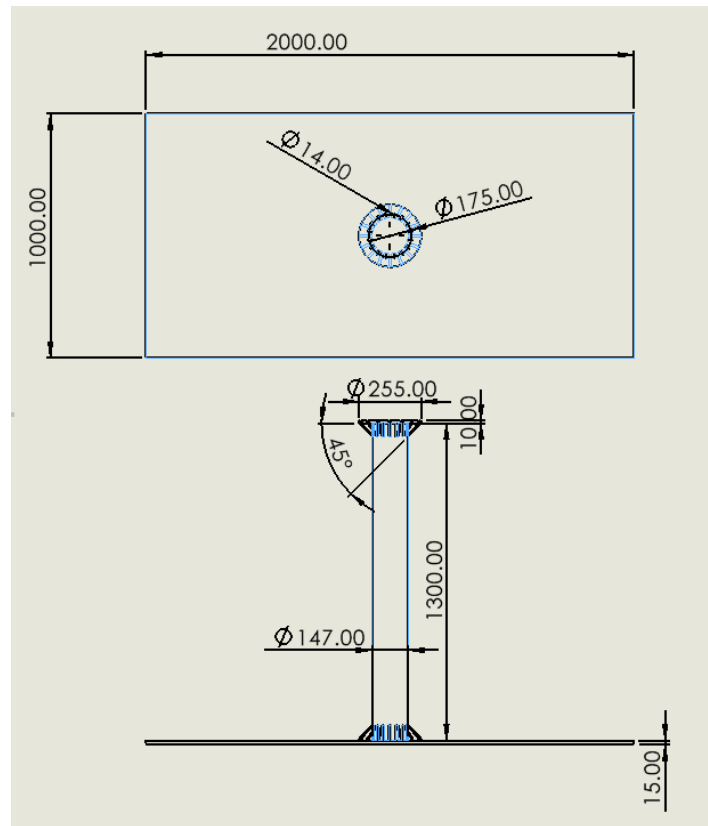


Figure 14: :2D drawing of vertical shaft attached to ground

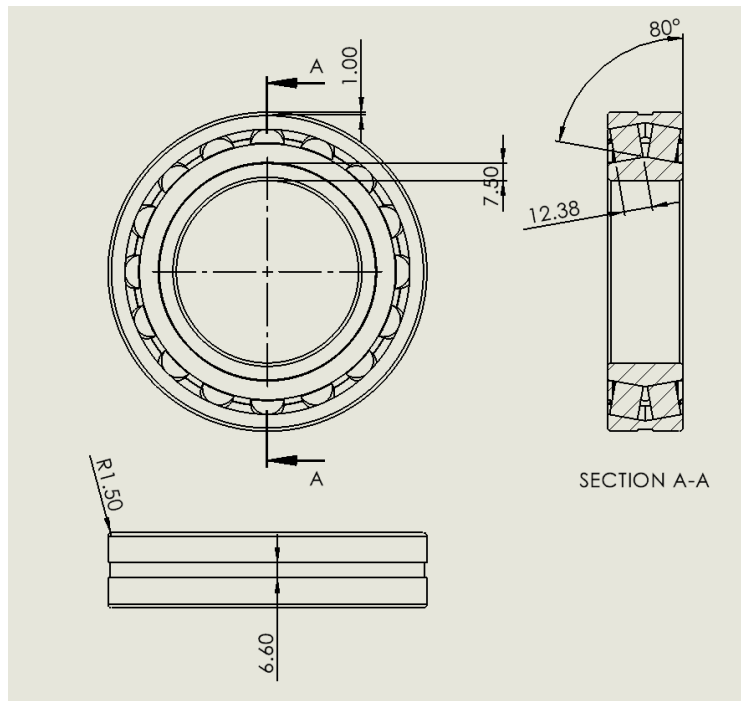


Figure 15: 2D drawing of spherical roller bearing